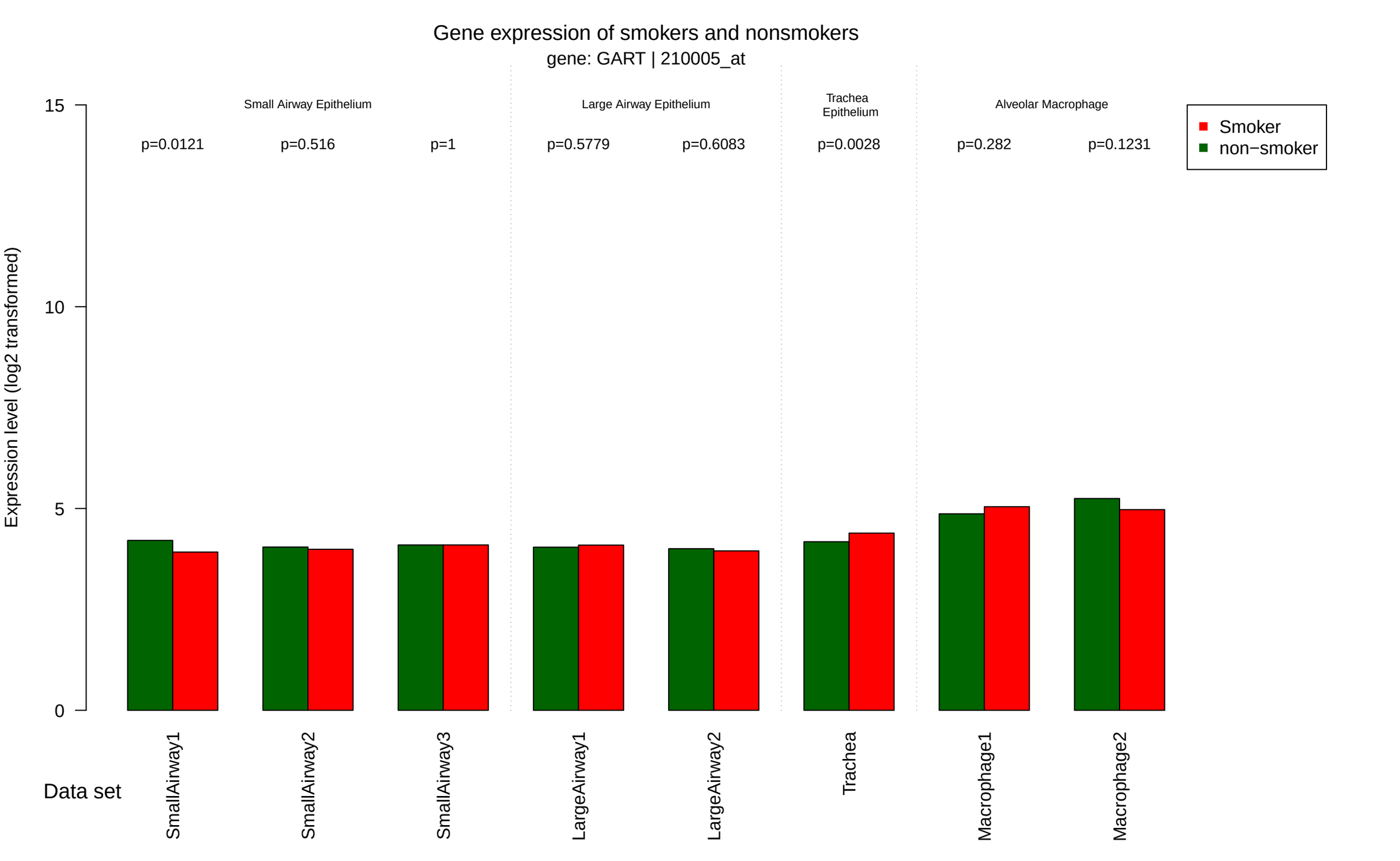
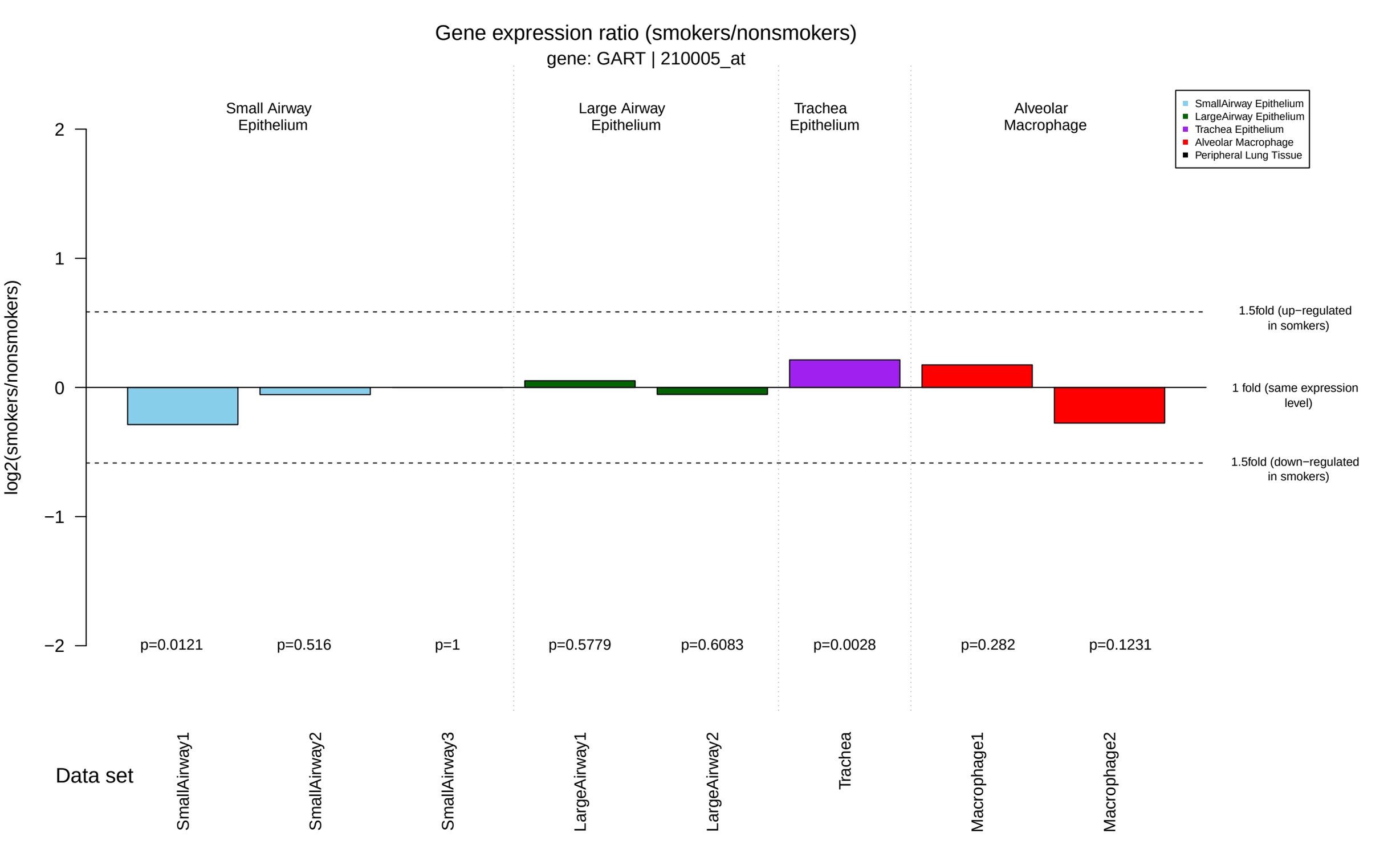
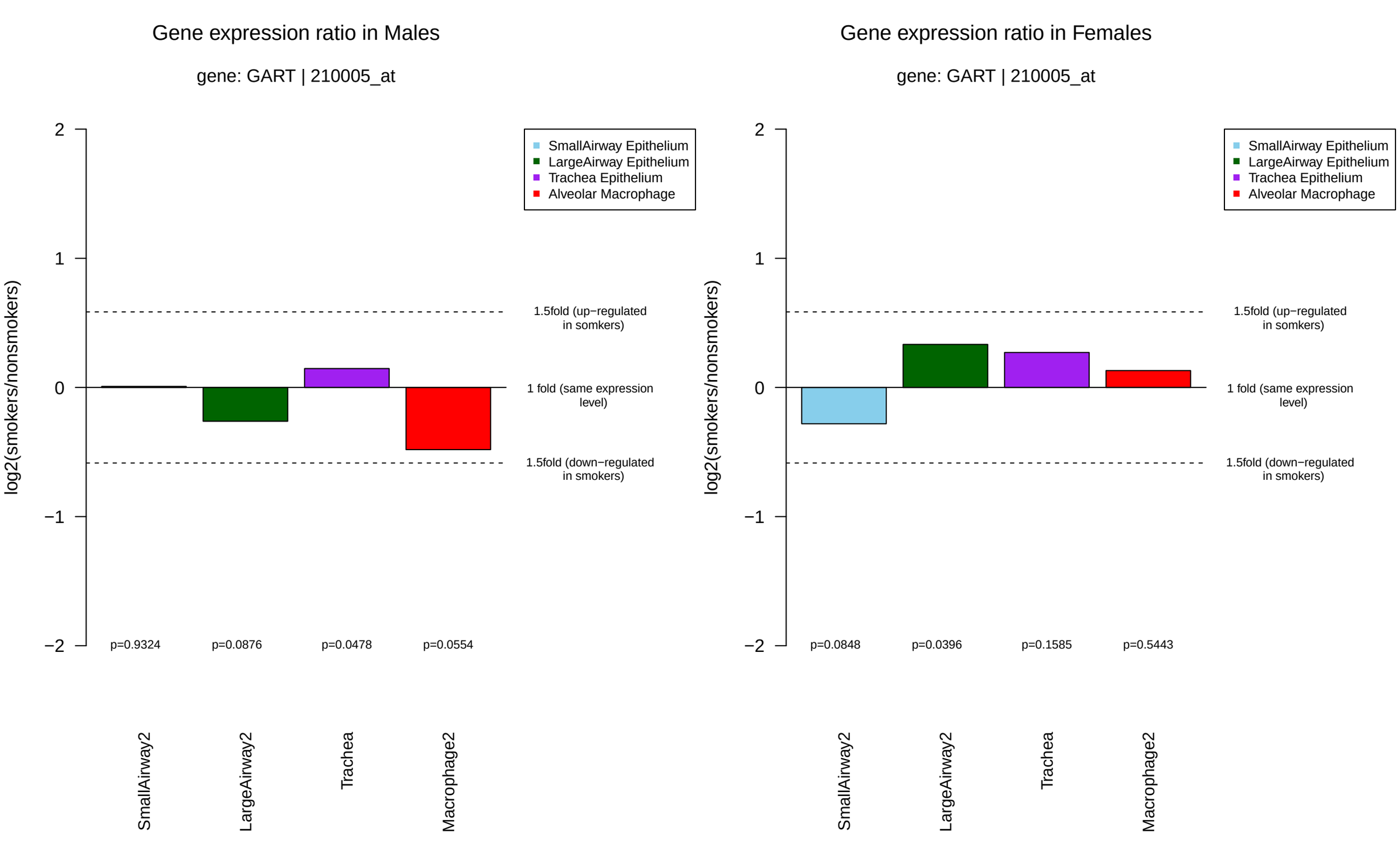
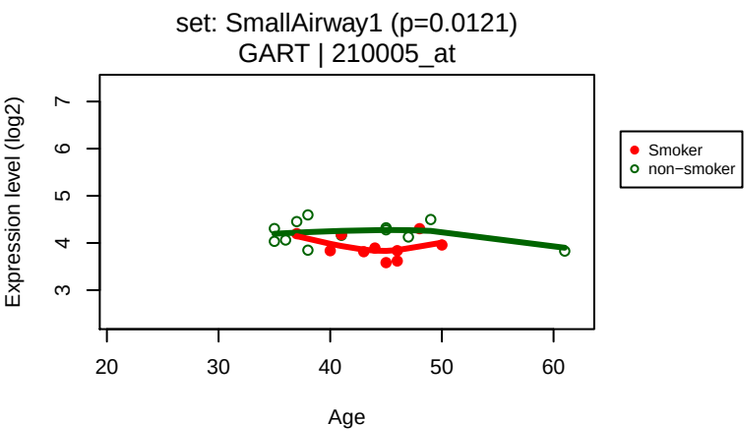
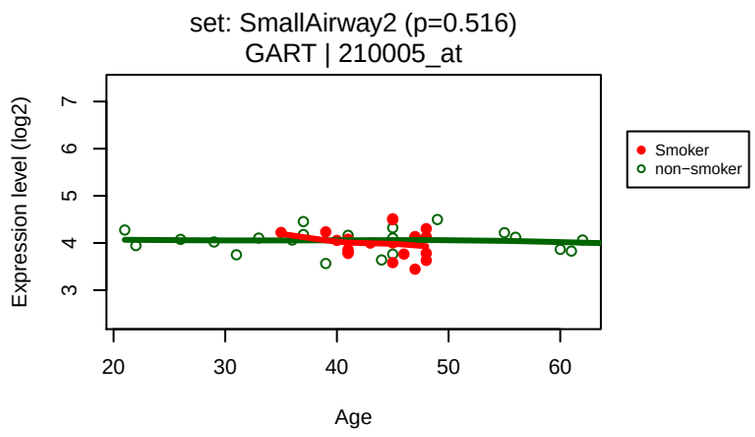
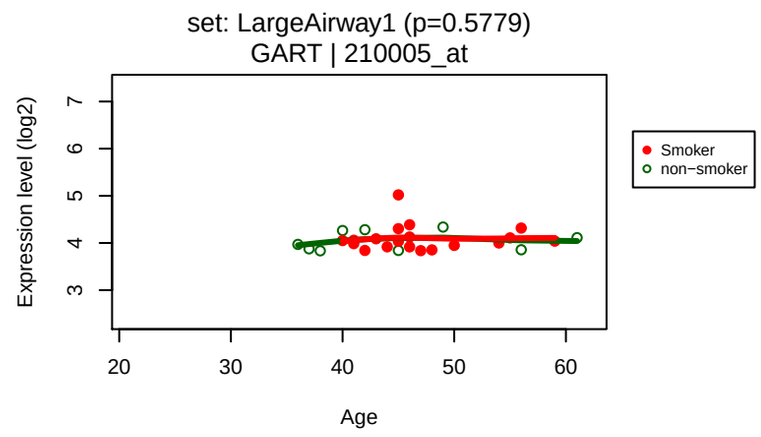

Gene expression of smokers and nonsmokers
gene: GART | 210005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0121
p=0.516
p=1
p=0.5779
p=0.6083
p=0.0028
p=0.282
p=0.1231
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

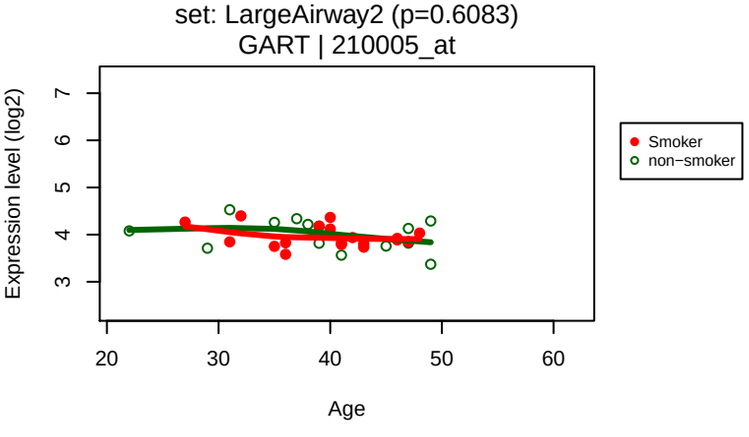

Gene expression ratio (smokers/nonsmokers)
gene: GART | 210005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0121
p=0.516
p=1
p=0.5779
p=0.6083
p=0.0028
p=0.282
p=0.1231
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

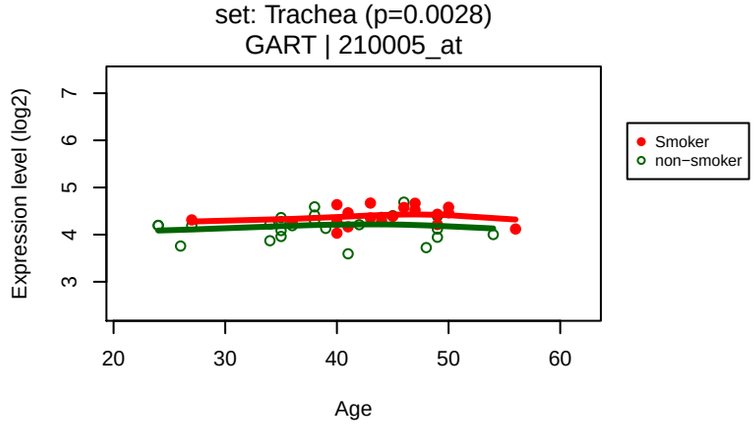

Gene expression ratio in Males
gene: GART | 210005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9324
p=0.0876
p=0.0478
p=0.0554
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GART | 210005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0848
p=0.0396
p=0.1585
p=0.5443
SmallAirway2
LargeAirway2
Trachea
Macrophage2

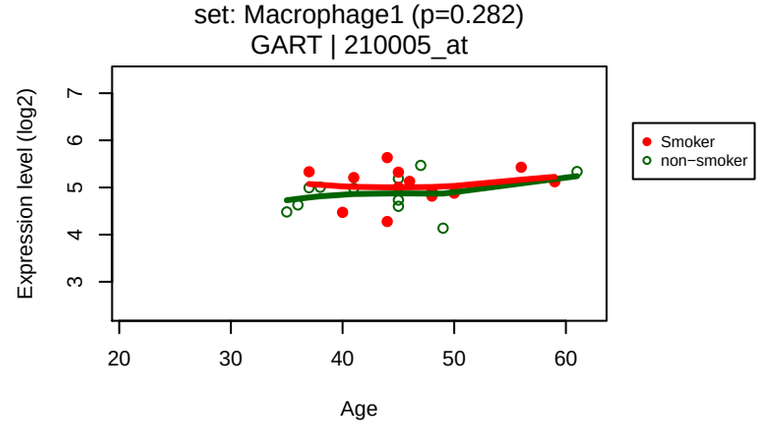

set: SmallAirway1 (p=0.0121)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker

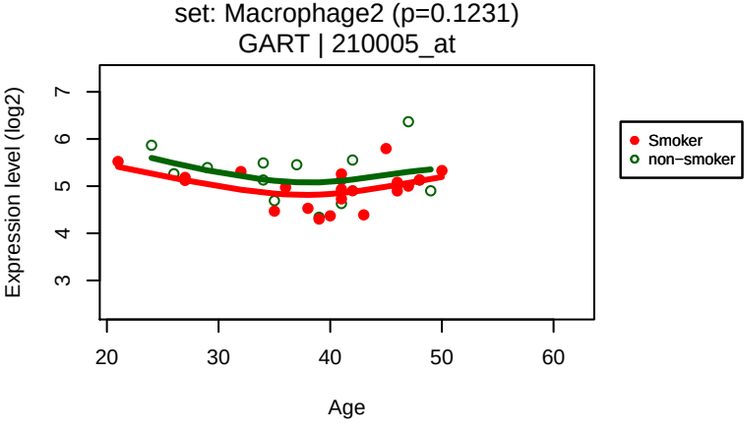

set: SmallAirway2 (p=0.516)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker

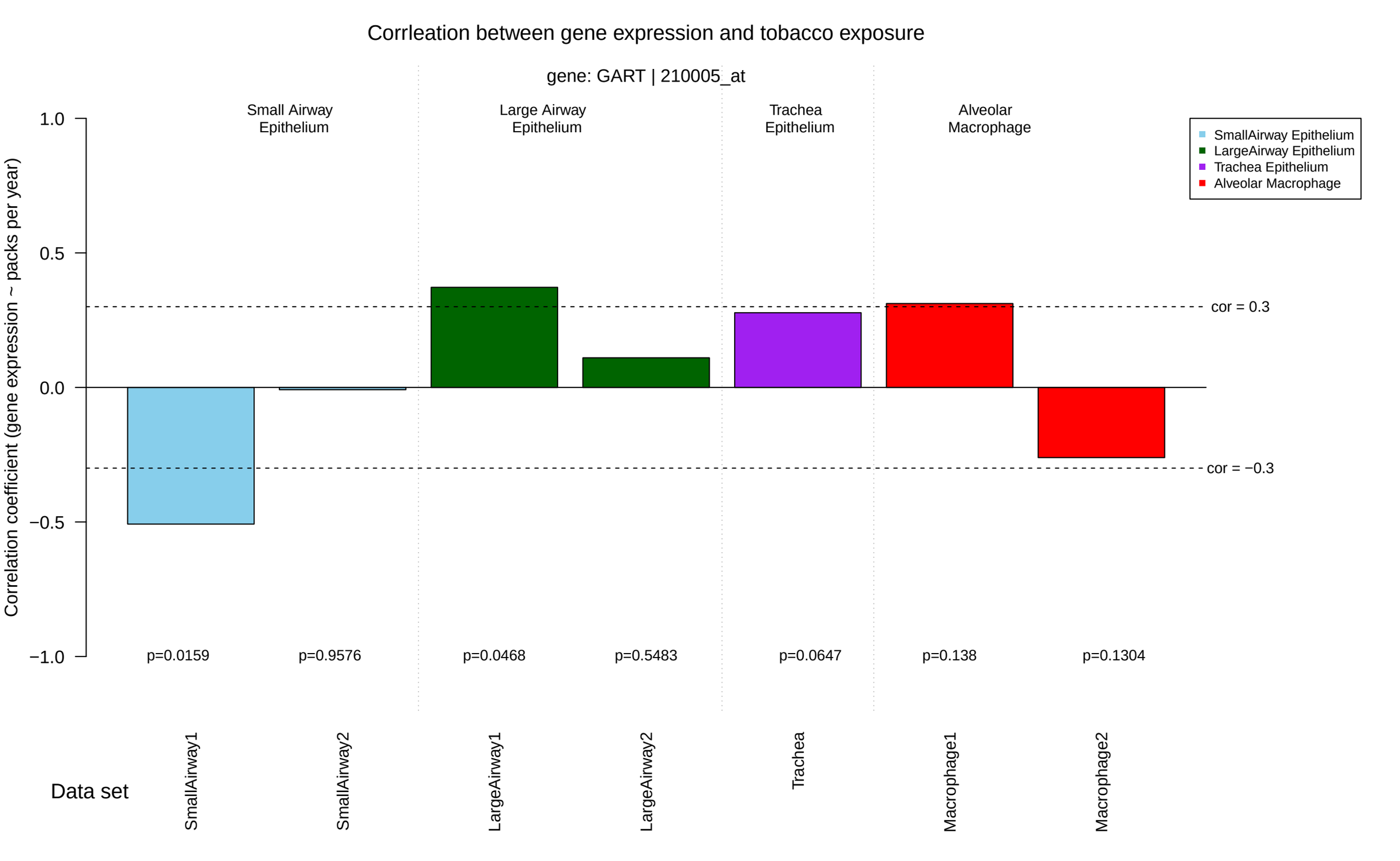

set: LargeAirway1 (p=0.5779)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6083)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0028)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.282)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1231)
GART | 210005_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GART | 210005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0159
p=0.9576
p=0.0468
p=0.5483
p=0.0647
p=0.138
p=0.1304
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2